

Laptop Price Prediction Using Ensemble Methods With Feature Engineering

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Abstract— The rapid advancement of technology has driven a significant surge in laptop demand, making it crucial to understand the factors that influence a laptop's performance and market value. While previous studies have successfully utilized machine learning for price prediction, a significant research gap remains regarding the depth of feature engineering applied, with many studies focusing primarily on algorithmic comparisons rather than data optimization. To address this limitation, this research presents an experimental comparative study aimed at building an accurate ensemble model to predict continuous laptop prices in Indian rupees using multi-feature structured data. Utilizing a dataset of 991 distinct laptop models, the study systematically tested 15 feature selection iterations across ensemble algorithms, including ExtraTrees, GradientBoosting, XGBoost, and RandomForest. The methodology strictly enforced an 80/20 train-test split and utilized 5-Fold Cross-Validation to ensure validity and prevent data leakage. The experimental results demonstrate that the Extremely Randomized Trees (ExtraTrees) algorithm, when paired with a Lasso reduced 20-variable feature subset ($\alpha = 10.0$), achieved the highest predictive performance. This optimal configuration yielded a Test R-squared (R^2) score of 0.9047, a Root Mean Squared Error (RMSE) of approximately 17,436 INR, and a Mean Absolute Error (MAE) of 9,991.77 INR. Ultimately, this study proves that rigorous feature subset optimization directly maximizes predictive accuracy, shifting the analytical focus toward a highly effective data-centric approach in consumer electronics valuation. Here are the following code for our research: https://colab.research.google.com/drive/136s1NI0xGY9efTzwrZsNsM_3YVeUJkZd?usp=sharing.

Keywords— Laptop Price Prediction, Ensemble Methods, Feature Engineering, ExtraTrees Algorithm, Machine Learning, Feature Selection.

I. INTRODUCTION

The rapid advancement of technology has driven a significant surge in laptop demand [1]. From everyday tasks to professional workflows, computers have transformed how people work, communicate, and obtain information [2]. As laptops become more widely used, consumers are confronted with a growing variety of models tailored to different requirements and applications. This increased dependence highlights how important it is to understand what influences a laptop's performance and value. Laptops have many factors that influence their specifications, such as CPU, RAM, storage type, and display technology [3]. Because laptops have various features that must be

considered before purchasing specific purposes [4], it can cause confusion for consumers when determining whether the laptop's price matches its specifications. Therefore, accurate prediction results can help laptop buyers choose the best laptop to buy based on its specifications [5]. The complexity of evaluating these diverse specifications can be objectively analyzed through a machine learning approach. Regression algorithms, such as Linear Regression, Random Forest, and XGBoost, have been successfully used in price prediction for several products based on historical data [7].

Despite these advancements, a significant research gap remains regarding the depth of feature engineering applied. Research by Qianlong, He developed a baseline prediction model [3], but the feature engineering was limited to basic categorical encoding and did not explore advanced techniques such as creating composite hardware features to capture complex interactions. Similarly, studies by Chaudhari and Patil [8] and Prajna et al. [9] demonstrated that Random Forest and multiple linear regression models could achieve high predictive performance, yet their primary focus was on algorithmic comparison or application development rather than sophisticated feature engineering. Even when advanced models like XGBoost were utilized and performed best, as shown by Kafabihi [1], they relied heavily on basic preprocessing without utilizing interaction terms or derived features. Therefore, a more detailed exploration of advanced feature engineering combined with ensemble models is needed to better understand how specification interactions influence laptop prices.

To address these limitations, this research aims to build an accurate ensemble model for predicting continuous laptop prices (in Indian rupees) based on multi-feature structured data. The study focuses on evaluating and comparing the effectiveness of various feature selection strategies (e.g., Recursive Feature Elimination, correlation-based filtering) to reduce data noise. By identifying the optimal feature subset that yields maximum prediction accuracy, the framework is objectively measured and validated through core error metrics including R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

This research contributes to a systematic framework for optimizing feature selection specifically tailored to ensemble models in consumer electronics price prediction. Unlike

previous studies [1, 3, 8] that compared algorithms superficially, this study demonstrates how rigorous feature subset optimization directly maximizes predictive accuracy. This shifts the analytical focus from basic algorithmic comparison to a data-centric approach, proving that precise feature selection in highly dimensional structured datasets is as critical as model complexity. The resulting optimized pipeline provides an actionable, replicable predictive model for retailers and consumers.

II. RELATED WORK

A. Machine Learning Applications in Consumer Electronics Pricing

The application of machine learning (ML) to price prediction in consumer electronics, particularly laptops, has gained substantial traction in recent literature. Foundation studies by [6] and [10] demonstrated that data-driven approaches effectively capture the relationships between technical specifications and market valuations. Subsequent research by [9] and [11] extended these methodologies, confirming the viability of ML models in structured data environments. Furthermore, recent studies highlight the continuous evolution of these predictive models through real-time web data analysis and data-driven predictive analytics to track dynamic market pricing [7].

B. Comparative Studies of Linear Regression and Ensemble-Based Regression Methods

Multiple studies have systematically compared single-model linear regression against ensemble-based regression techniques. Research by [5] and [12] evaluated linear baselines against ensemble techniques like Random Forest, finding that ensemble methods consistently achieved lower error rates due to their ability to model complex feature interactions. In focused comparisons, [8] and [13] demonstrated that Decision Trees and Random Forests frequently outperform multiple linear regression when feature engineering adequately captures underlying price determinants. Gradient boosting algorithms have also emerged as highly effective; [2] and [4] found that Histogram Gradient Boosting and XGBoost substantially outperformed linear baselines. These algorithmic trends are supported by related hardware pricing research, such as [14], which observed the superiority of XGBoost when predicting GPU prices. Additionally, advanced models like LSTM have been benchmarked against classical regression in specific niches, such as gaming laptops [15].

C. Feature Engineering Strategies

The critical role of feature engineering in predictive performance is well-documented. He [3] demonstrated that systematic categorical encoding and transformation of hardware specifications substantially improves model fit compared to using raw data. Further explorations by [1] and [16] into the specific influencing factors of laptop prices concluded that domain-specific transformations such as aggregating secondary storage configurations or grouping processor tiers are vital for accurately mapping objective hardware metrics to subjective market values in structured datasets.

III. RESEARCH METHODOLOGY

A. Research Method Used

The method employed is an experimental comparative study focusing entirely on predictive performance optimization. The experiment systematically tests various

feature selection iterations on an ensemble model architecture (e.g., Random Forest or XGBoost) to identify the specific feature subset that yields the lowest error rates and the highest explained variance (R^2).

B. Dataset and Data Collection Strategy

This study relies on secondary data collection using the publicly available Brand Laptops Dataset from Kaggle. The dataset contains specifications for 991 distinct laptop models and includes 22 raw attributes, covering brand, processor details, RAM, storage, GPU, display size, operating system, and the target variable, price (in Indian rupees). The high dimensionality of these technical features provides an ideal testing ground to demonstrate how eliminating redundant or irrelevant variables reduces noise in ensemble models.

C. Experimental Procedure

The experimental pipeline is structured into the following systematic steps:

1. *Data Loading & Preprocessing*: Import the raw dataset, handle missing values, and transform the data. Categorical variables, such as processor hierarchy and operating system, are converted into discrete numerical or binary representations.
2. *Train-Test Splitting*: Isolate the data into a Training Set (80%) and a Testing Set (20%). The testing set is strictly locked and will not be exposed to the model until the final evaluation phase.
3. *Feature Standardization*: Scale numerical features using a StandardScaler. This ensures that features with inherently large numerical ranges (e.g., storage capacity) do not distort the computational process.
4. *Feature Selection Experimentation*: Apply distinct feature selection techniques independently on the training set. This creates competing “feature subsets” (e.g., Subset 1 using all features as a baseline, Subset 2 using correlation-based filtering, Subset 3 using Recursive Feature Elimination).
5. *Ensemble Model Training*: Train identical instances of the chosen ensemble algorithm on each of the competing feature subsets using the training data.
6. *Performance Evaluation*: Generate price predictions on the unseen testing set for each model variant. Measure and compare their performance strictly using objective error metrics.

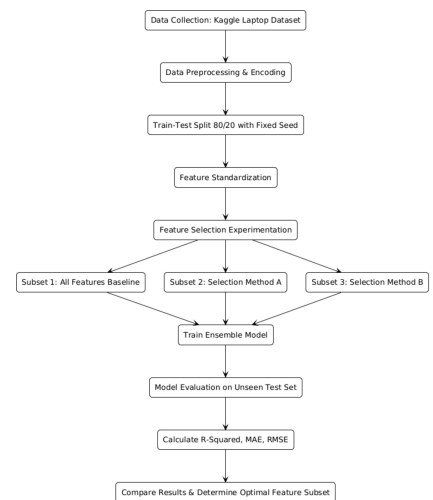


Fig. 1. Illustrations of the Experimental Workflow

D. Ensuring Validity and Reproducibility

Internal validity is enforced by preventing data leakage; the testing set is completely isolated from both feature selection process and model training. Furthermore, K-Fold Cross-Validation is applied during the feature selection execution within the training set. This guarantees that the selected feature combinations represent stable predictive power and are not merely illusions caused by overfitting to a specific data sample. Total system reproducibility is guaranteed by utilizing an open-source public dataset. Every computational randomization process—including the train-test split, the distribution of folds in cross-validation, and the internal tree initialization of the ensemble algorithm—utilizes a fixed random seed. Any independent researcher executing this pipeline will reproduce the exact same error values (MAE, RMSE) down to the decimal point.

E. Feature Engineering

The methodology employed to transform raw unstructured hardware specifications into a computable dataset was centered on ensuring consistency and statistical neutrality. To mitigate the biases inherent in raw text entries we performed a systematic extraction of concrete numerical values. Memory capacities were standardized as Storage_Primary_GB while processor frequencies were converted into a continuous CPU_GHz format and display specifications were normalized into a singular Resolution_Pixels metric.

Beyond basic extraction we engineered composite variables to better represent the technical synergy between hardware components. The Hardware_Index was developed to quantify the interaction between RAM and processing speed while the Performance_Density metric was formulated to evaluate processing capability relative to the device's physical weight. Following the resolution of categorical labels through encoding all input features underwent standard scaling. This step was critical for ensuring that variables with large numerical ranges such as storage capacity and pixel count did not disproportionately influence the model's weight distribution and it maintained a balanced learning environment across the entire feature space.

IV. RESULTS

A. Evaluation of Feature Subset Scenarios

Following the data preparation, we tested several ensemble algorithms, specifically ExtraTrees, GradientBoosting, XGBoost, and RandomForest across 15 distinct feature combinations. To guarantee that our findings represent true predictive capability rather than a fortunate train-test split, we anchored the validation process with a 5-Fold Cross-Validation (CV) strictly on the training set. The final performance was then measured against an isolated 20% holdout test set.

The following table summarizes the top ten configurations based on their final R-squared (R^2) test scores:

TABLE I. Summary of Configurations Based on Test R-Squared Scores

Algorithm	Feature Subset	Feature Count	Test R^2	RMSE	MAE
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Extra Trees	Lasso (alpha = 10.0)	20	0.9047	17435.97	9991.77
Extra Trees	Baseline (All Features)	21	0.9008	17782.40	10223.29
Extra Trees	Lasso (alpha = 1.0)	21	0.9008	17782.40	10223.29
Extra Trees	Lasso (alpha = 0.1)	21	0.9008	17782.40	10223.29
Extra Trees	Variance Threshold	21	0.9008	17782.40	10223.29
Extra Trees	KBest MI (Top 10)	10	0.8788	19658.45	11778.61
Gradient Boosting	Baseline (All Features)	21	0.8709	20289.13	11718.35
Gradient Boosting	Variance Threshold	21	0.8709	20289.13	11718.35
Gradient Boosting	Lasso (alpha = 1.0)	21	0.8709	20289.13	11718.35
Gradient Boosting	Lasso (alpha = 0.1)	21	0.8709	20289.13	11718.35
Extra Trees	Percentile 75	15	0.8670	20590.84	12051.93
Extra Trees	KBest_f_regression_top_15	15	0.8670	20590.84	12051.93
Gradient Boosting	Lasso (alpha = 10.0)	20	0.8658	20686.29	11719.48
XGBoost	Lasso (alpha = 10.0)	20	0.8608	21065.88	10847.44
Gradient Boosting	KBest MI (Top 10)	10	0.8606	21081.25	12297.08
Gradient Boosting	RFE RF (Top 10)	10	0.8571	21349.85	12091.53
Gradient Boosting	Percentile 75	15	0.8559	21435.60	12574.45
Random Forest	KBest MI (Top 5)	5	0.8559	21438.24	12288.47

Gradient Boosting	KBest_f_regression_top_15	15	0.8559	21435.60	12574.45
XGBoost	KBest MI (Top 10)	10	0.8535	21610.93	11637.43
XGBoost	Lasso (alpha = 0.1)	21	0.848	22017.94	11330.32
XGBoost	Lasso (alpha = 1.0)	21	0.848	22017.94	11330.32
XGBoost	Baseline (All Features)	21	0.848	22017.94	11330.32
XGBoost	Variance Threshold	21	0.848	22017.94	11330.32
Extra Trees	KBest MI (Top 5)	5	0.8452	22215.46	12324.95
Extra Trees	RFE RF (Top 10)	10	0.8427	22398.29	12970.8
Random Forest	Lasso (alpha = 10.0)	20	0.8427	22399.78	11772.28
XGBoost	Percentile 50	10	0.8424	22419.63	12859.4
XGBoost	KBest_f_regression_top_10	10	0.8424	22419.63	12859.4
Random Forest	Lasso (alpha = 1.0)	21	0.8420	22444.22	11839.26
Random Forest	Variance Threshold	21	0.8420	22444.22	11839.26
Random Forest	Lasso (alpha = 0.1)	21	0.8420	22444.22	11839.26
Random Forest	Baseline (All Features)	21	0.8420	22444.22	11839.26
XGBoost	Percentile 75	15	0.8417	22469.48	12262.26
XGBoost	KBest_f_regression_top_15	15	0.8417	22469.48	12262.26

Gradient Boosting	Percentile 50	10	0.8416	22472.95	13008.22
Gradient Boosting	KBest_f_regression_top_10	10	0.8416	22472.95	13008.22
XGBoost	RFE RF (Top 10)	10	0.8374	22769.45	12059.34
XGBoost	RFE RF (Top 5)	5	0.8352	22924.69	13653.18
Extra Trees	KBest_f_regression_top_10	10	0.8337	23025.54	14096.28
Extra Trees	Percentile 50	10	0.8337	23025.54	14096.28
Gradient Boosting	RFE RF (Top 5)	5	0.8326	23104.58	13699.41
XGBoost	KBest MI (Top 5)	5	0.8270	23486.19	12111.45
Random Forest	RFE RF (Top 10)	10	0.8264	23529.21	13116.48
Extra Trees	RFE RF (Top 5)	5	0.8252	23613.26	14832.28
Random Forest	Percentile 75	15	0.8220	23825.54	12828.68
Random Forest	KBest_f_regression_top_15	15	0.8220	23825.54	12828.68
Gradient Boosting	Percentile 25	5	0.8150	24289.92	14778.79
Gradient Boosting	KBest_f_regression_top_5	5	0.8150	24289.92	14778.79
Random Forest	KBest MI (Top 10)	10	0.8127	24438.46	12817.26
XGBoost	Percentile 25	5	0.8026	25092.85	15185.25
XGBoost	KBest_f_regression_top_5	5	0.8026	25092.85	15185.25

Random Forest	Percentile 50	10	0.7998	25270.32	13925.71
Random Forest	KBest_f_regression_top_10	10	0.7998	25270.32	13925.71
Gradient Boosting	KBest MI (Top 5)	5	0.7958	25518.61	14293.74
Random Forest	RFE RF (Top 5)	5	0.7799	26496.02	14966.47
Random Forest	KBest_f_regression_top_5	5	0.7634	27470.40	16107.75
Random Forest	Percentile 25	5	0.7634	27470.40	16107.75
Extra Trees	Percentile 25	5	0.7507	28195.64	16743.69
Extra Trees	KBest_f_regression_top_5	5	0.7507	28195.64	16743.69

B. Analysis of Model Performance and Feature Significance

Our experimental findings demonstrate that the ExtraTrees algorithm consistently outperforms the gradient boosting variants typically favored in existing literature for physical asset pricing. The model achieved its peak performance when paired with a Lasso-reduced feature set, yielding a test R^2 of 0.9047 and an RMSE of approximately 17,436 INR. The algorithm's inherent extreme randomization during node splits proves uniquely effective at filtering out the high local variance and noise characteristic of modern consumer electronics pricing.

Structural stability is well-supported by our cross-validation results. The top configuration maintained a mean CV R^2 of 0.8843, with a notably low standard deviation ($SD = 0.0217$) across five folds, indicating that the architecture avoids localized overfitting and effectively captures broader market dynamics. Peak optimization was achieved by applying a Lasso penalty of 10.0 to isolate a 20-variable subset, which pushed the MAE down to 9,991.77 INR. This confirms that pruning statistical noise is essential for sharpening an ensemble's predictive accuracy.

Furthermore, the framework shows remarkable resilience under aggressive dimensionality constraints. Even when restricted to the top 10 features via Mutual Information filtering, the model maintained a robust R^2 of 0.8788. This underscores a key takeaway in the retail hardware landscape: while mapping a broad spectrum of technical data is necessary to surpass the 90% accuracy threshold, the fundamental valuation of modern laptops is driven primarily by a concentrated core of roughly ten macro-specifications.

V. CONCLUSION

This study concludes that rigorous feature subset optimization can directly maximize the accuracy of laptop price prediction, shifting the analytical focus to a highly effective data-centric approach in consumer electronics pricing. Through systematic testing of 15 iterations of feature selection across various ensemble algorithms, the Extremely Randomized Trees (ExtraTrees) model paired with a 20-feature subset reduced using Lasso ($\alpha=10.0$) was shown to yield the highest predictive performance. This optimal configuration achieved a Test R-squared (R^2) value of 0.9047, a Root Mean Squared Error (RMSE) of approximately 17,436 INR, and a Mean Absolute Error (MAE) of 9,991.77 INR.

The results also show that the ExtraTrees algorithm consistently outperforms gradient boosting variants due to its highly effective ability to filter out the high local variance and statistical noise characteristic of modern electronics pricing. This statistical noise reduction proves to be essential for sharpening the prediction accuracy of an ensemble model. Finally, the framework demonstrates that while mapping a broad spectrum of technical data is required to surpass the 90% accuracy threshold, the underlying price valuation of a modern laptop is primarily driven by a core group concentrated around ten macro-specifications.

VI. AUTHOR CONTRIBUTION

Ezell Joseph Kurniawan: Methodology, Software, Data Curation, Writing – Original Draft (Related Work section), Writing – Review & Editing (Research Methodology, paragraphs 3–6; Conclusion, paragraph 6)

Jason Sebastian Sugianto: Conceptualization, Validation, Resources, Writing – Review & Editing (Related Work, paragraph 2; Research Methodology, paragraphs 2–4)

Muhammad Ridho Prakoso: Conceptualization, Writing – Original Draft (Abstract, Introduction, Related Work; Research Methodology, paragraphs 2–4, including translation), Writing – Review & Editing (Research Methodology, paragraphs 2–4)

Henry Lucky: Supervision

Rifqi Alfinnur Charisma: Supervision

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